The documentation and process conversion measures necessary to comply with this revision shall be completed by 9 March 98

INCH POUND

MIL-PRF-19500/391D
9 December 1997
SUPERSEDING
MIL-S-19500/391C
29 October 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES 2N3019, 2N3019S, 2N3057A, 2N3700, AND 2N3700UB
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low-power transistors. Four levels of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1(similar to T0-39), 3 (similar to T0-46), 4 (similar to T0-18), and 6 (surface mount).

1.3 Maximum ratings.

Type	P_T		V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_J and T_{STG}
	T_A = +25°C 1/	T_C = +25°C 2/					
	W	W	V dc	V dc	V dc	A dc	°C
2N3019	0.8	5	140	7	80	1	-55 to +175
2N3019S	0.8	5	140	7	80	1	-55 to +175
2N3057A	0.4	1.8	140	7	80	1	-55 to +175
2N3700	0.5	1.8	140	7	80	1	-55 to +175
2N3700UB	0.4	1.16	140	7	80	1	-55 to +175

1/ Derate linearly at 4.6 mW/°C for types 2N3019 and 2N3019S; 2.3 mW/°C for type 2N3057A; and 2.85 mW/°C for type 2N3700; 6.6 mW/°C for type 2N3700UB for $T_A \geq +25°C$.
2/ Derate linearly at 28.6 mW/°C for types 2N3019 and 2N3019S; 10.3 mW/°C for types 2N3057A, 2N3700, and 2N3700UB for $T_C \geq +25°C$.

1.4 Primary electrical characteristics.

Limits	h_{FE1}	h_{FE2}	h_{FE3} 1/	$\|h_{fe}\|$	$V_{CE(sat)1}$ 1/
	V_{CE} = 10 V dc I_C = 150 mA dc	V_{CE} = 10 V dc I_C = 0.1 mA dc	V_{CE} = 10 V dc I_C = 10 mA dc	V_{CE} = 10 V dc I_C = 50 mA dc f = 20 MHz	I_C = 150 mA dc I_B = 15 mA dc
					V dc
Min	100	50	90	5	
Max	300	200		20	0.20

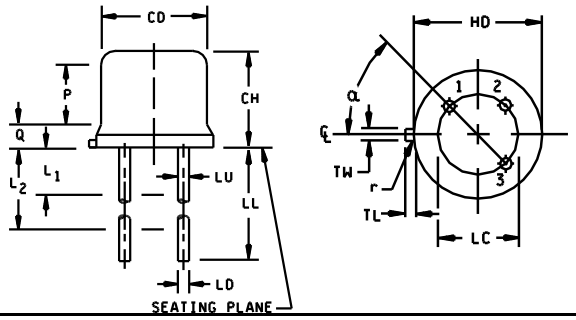
1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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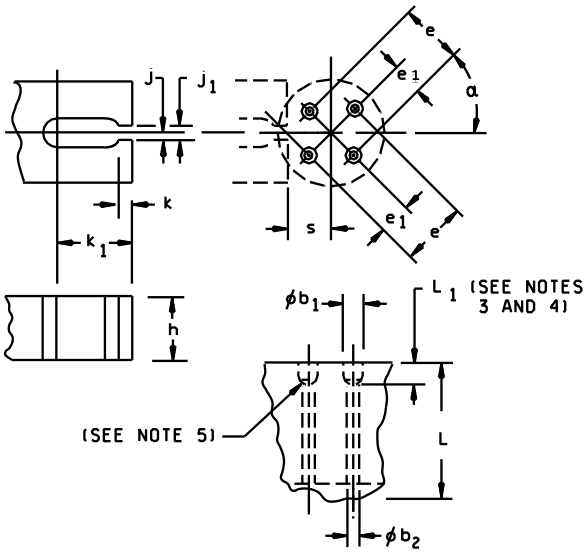


Dimensions					
Symbol	Inches		Millimeters		Notes
CD	0.305	0.335	7.75	8.51	
CH	0.240	0.260	6.10	6.60	
HD	0.335	0.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	0.016	0.021	0.41	0.53	7, 8
LL	See notes				7, 8, 11,12
LU	0.016	0.019	0.41	0.48	7, 8
L1		0.050		1.27	7, 8
L2	0.250		6.35		7, 8
P	0.100		2.54		5
Q		0.050		1.27	4
r		0.010		0.25	10
TL	0.029	0.045	0.74	1.14	3
TW	0.028	0.034	0.71	0.86	2
α	45° TP		45° TP		6

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. For transistor type 2N3019, dimension LL shall be 1.5 inches (38.1) mm minimum and 1.75 inches (44.4 mm) maximum.
12. For transistor type 2N3019S, dimension LL shall be 0.5 inches (12.7) mm minimum and 0.75 inches (19.0 mm) maximum.
13. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
14. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions for device types 2N3019 and 2N3019S

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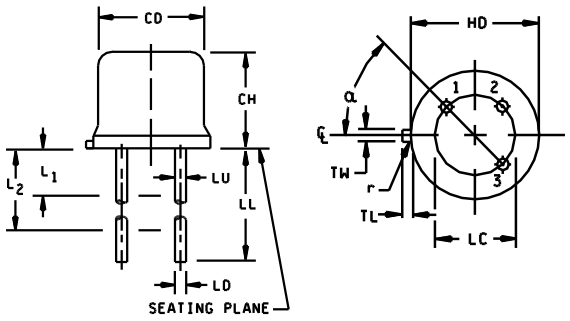
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕb_1	.0595	.0605	1.511	1.537	
ϕb_2	.0325	.0335	0.826	0.851	
e	.1995	.2005	5.067	5.093	
e_1	.0995	.1005	2.527	2.553	
h	.150 Nominal		3.81 Nominal		
j	.0175	.0180	0.444	0.457	
j_1	.0350	.0355	0.889	0.902	
k	.009	.011	0.23	0.28	
k_1	.125 Nominal		3.18 Nominal		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	2
α	44.90°	45.10°	44.90°	45.10°	

NOTES:

1. Dimensions are in inches. Metric equivalents are given for general information only.
2. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
3. Gauging procedure. The device being measured shall be inserted until its seating plane is .125 ±.010 inch (3.18 ±0.25 mm) from the seating surface of the gauge. A force of 8 ±.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed), the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible.
4. Gauging plane.
5. Drill angle.

FIGURE 2. Gauge for lead and tab location for device types 2N3019 and 2N3019S.

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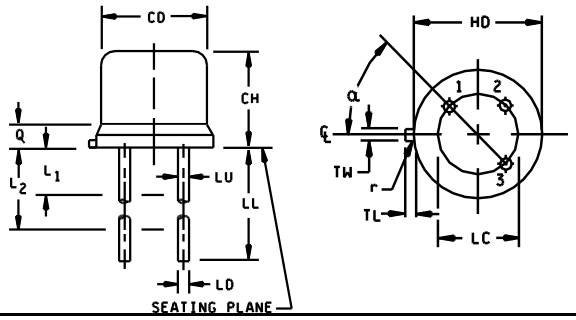


Dimensions					
Symbol	Inches		Millimeters		Notes
CD	0.178	0.195	4.52	4.95	
CH	0.065	0.085	1.65	2.16	
HD	0.209	0.230	5.31	5.84	
LC	.100 TP		2.54 TP		4
LD	0.016	0.021	0.41	0.53	5, 6
LL	.500	.750	12.7	19.05	5, 6
LU	0.016	0.019	0.41	0.48	5, 6
L1		0.050		1.27	5, 6
L2	0.250		6.35		5, 6
r		0.010		0.25	8
TL	0.028	0.048	0.71	1.22	2, 3
TW	0.036	0.046	0.91	1.17	2
α	45° TP		45° TP		4

NOTES:
1. Dimension are in inches. Metric equivalents are given for general information only.
2. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
3. Dimension TL measured from maximum HD.
4. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 5.
5. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
6. All three leads.
7. The collector shall be internally connected to the case.
8. Dimension r (radius) applies to both inside corners of tab.
9. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
10. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 3. Physical dimensions for 2N3057A.

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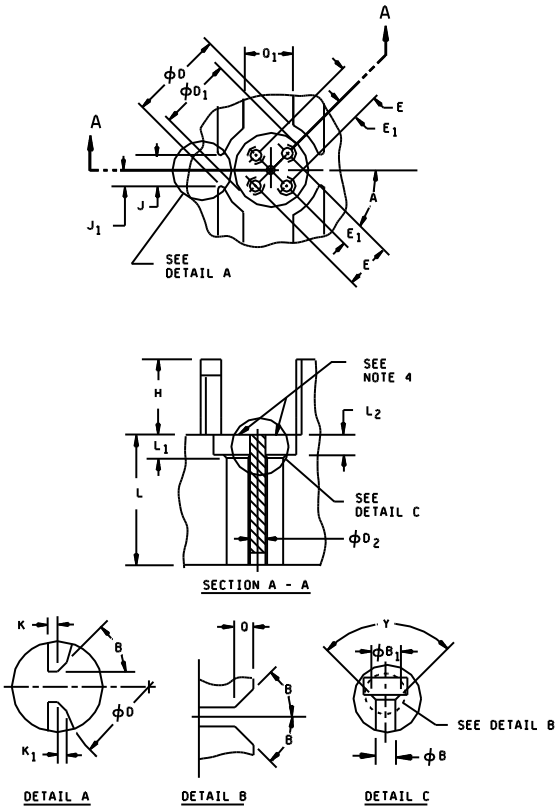
Dimensions					
Symbol	Inches		Millimeters		Notes
CD	0.178	0.195	4.52	4.95	
CH	0.170	0.210	4.32	5.33	
HD	0.209	0.230	5.31	5.84	
LC	.100 TP		2.54 TP		5
LD	0.016	0.021	0.41	0.53	6, 7
LL	.500	.750	12.70	19.05	6, 7
LU	0.016	0.019	0.41	0.48	6, 7
L1		0.050		1.27	6, 7
L2	0.250		6.35		6, 7
Q		0.040		1.02	4
r		0.010		0.25	9
TL	0.028	0.048	0.71	1.22	2, 3
TW	0.036	0.046	0.91	1.17	2
α	45° TP		45° TP		5

NOTES:
1. Dimension are in inches. Metric equivalents are given for general information only.
2. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
3. Dimension TL measured from maximum HD.
4. Body contour optional within zone defined by HD, CD, and Q.
5. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 5.
6. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
7. All three leads.
8. The collector shall be internally connected to the case.
9. Dimension r (radius) applies to both inside corners of tab.
10. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
11. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 4. Physical dimensions for type 2N3700.

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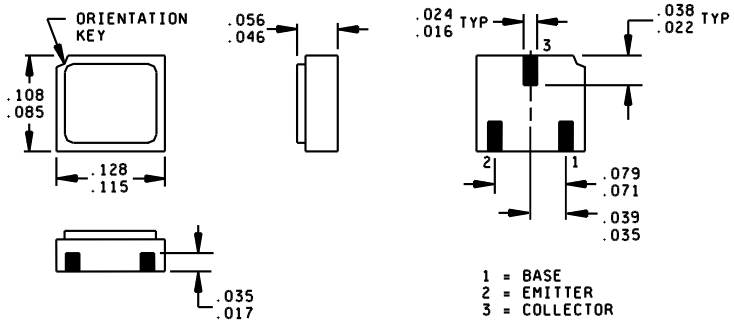
Ltr	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
ϕB	.8255	.8509	.0325	.0335	5
ϕB_1	1.09 Nom		.040 Nom		5
ϕD	5.867	5.880	.2310	.2315	
ϕD_1	4.04	4.09	.159	.161	
ϕD_2	1.02 Nom		.040 Nom		6
E	2.527	2.553	.0995	.1005	
E_1	1.257	1.283	.0495	.0505	
H	3.68	3.94	.145	.155	
J	1.194	1.207	.0470	.0475	
J_1	.597	.622	.0235	.0245	
K	.229	.279	.009	.011	
K_1	.127 Nom		.005 Nom		
L	9.45	9.60	.372	.378	
L_1	1.37	1.40	.054	.055	
L_2	1.09 Nom		.043 Nom		
Q	1.02 Nom		.040 Nom		
Q_1	3.12	3.23	.123	.127	
α	---	---	44.90°	45.10°	
B	---		45° Nom		
Y	---		90° Nom		



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) ±.010 inch (0.254 mm) from the seating surface of the gauge. A force of 8 ±.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
4. These surfaces to be parallel and in same plane within ±.001 inch (0.025 mm).
5. Four holes.
6. Pressed in.

FIGURE 5. Gauge for lead and tab location for device types 2N3057A and 2N3700.

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Inches	mm	Inches	mm
.016	0.41	.056	1.42
.017	0.43	.071	1.81
.022	0.56	.079	2.01
.024	0.61	.085	2.41
.035	0.89	.108	2.74
.038	0.96	.115	2.82
.039	0.99	.128	3.25
.046	0.97		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 6. Physical dimensions, surface mount (2N3700UB).

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federaland military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1 (2N3019 and 2N3019S, similar to T0-39), 3 (2N3700, similar to T0-46), 4 (2N3057, similar to T0-18), and 6 (2N3700UB, surface mount) herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANTX, JANTXV, and JANS only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{CES1} and h_{FE1}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CES1}; h_{FE1}; ΔI_{CES1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15%.	I_{CES1} and h_{FE1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 & 3 of table I herein; ΔI_{CES1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15%.	Subgroup 2 of table I herein; ΔI_{CES1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15%.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750 (see 4.5).

JANS level - V_{CB} = 10 V dc, P_T = 800 mW for 2N3019 and 2N3019S, 400 mW for 2N3057A, 2N3700UB, and 500 mW for 2N3700.
JANTX and JANTXV levels - V_{CB} = 60 V dc, P_T = 800 mW for 2N3019 and 2N3019S, 400 mW for 2N3057A, 2N3700UB, and 500 mW for 2N3700.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal impedance ($Z_{\Theta JX}$ measurements). The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current - - - - - - - - - - - 5 mA.
b. I_H forward heating current - - - - - - - - 200 mA (min).
c. t_H heating time - - - - - - - - - - - - - - 25 - 30 ms.
d. t_{md} measurement delay time - - - - - - - - 60 µs max.
e. V_{CE} collector-emitter voltage - - - - - - - 10 V dc minimum

The maximum limit for $Z_{\Theta JX}$ under these test conditions are $Z_{\Theta JX}$ (max) = 72° C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements shall be in accordance with the steps of table II herein as specified in the notes for table II.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
4	1037	V_{CB} = 10 V dc; 2N3019 and 2N3019S - P_T = 800 mW; 2N3057A, 2N3700UB - P_T = 400 mW; 2N3700 - P_T = 500 mW at T_A = Room ambient as defined in MIL-STD-750, $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles (see 4.5.1). No heat sink or forced-air cooling on the devices shall be permitted.
5	1027	V_{CB} = 10 V dc; T_A = 125°C ±25°C for 96 hours; 2N3019 and 2N3019S - P_T = 800 mW; 2N3057A, 2N3700UB - P_T = 400 mW; 2N3700 - P_T = 500 mW at T_A = 100°C or adjusted as required by the chosen T_A to give an average lot T_J = 275°C. Marking legibility requirements shall not apply (see 4.5.1).
5	2037	Test condition A.
6		Not applicable.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	HTRB, test condition A, T_A = 150°C, V_{CB} = 112 V dc, 340 hours, n = 22 devices, c = 0.
2	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 V dc, P_T = 800 mW for 2N3019 and 2N3019S; P_T = 400 mW for 2N3057A and 2N3700UB; P_T = 500 mW for 2N3700; at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on the devices shall be permitted.
	or	n = 45 devices, Acc = 0
	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles, $\Delta T_J \geq +100°C$; forced air cooling allowed on cooling cycle only. n = 45 devices, Acc = 0.
3	2037	Bond strength, n = 11 wires, c = 0.
4	1032	High-Temperature life (non-operating), T_A = +200°C n = 22, c = 0

1/ Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. Samples shall be selected from each wafer lot from a die fabrication line that the manufacturer chooses to "certify" for the die design used in products manufactured to this specification.

b. Must be chosen from an inspection lot that has been submitted to and passed group A conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B3 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.2.4 Product acceptable for delivery. The product shall be acceptable for delivery only after the successful completion of all group B testing and shall be comprised of die meeting the following requirements:

a. Manufactured on the same certified die fabrication line as the sample selected for 4.4.2.3.

b. The die lot was manufactured (using start or completion dates, as the manufacturer selects to consistently use) within three months following the wafer lot used for sample selection in paragraph 4.4.2.3. Group B shall not be required more than once for any single wafer lot.

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4.4.3 Group C inspection, JANS. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 herein (JANS). See 4.4.3.2 for JAN, JANTX, and JANTXV group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with the steps of table II herein as specified in the notes for table II.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UB devices.
C6	1026	conditions: 1,000 hours at V_{CB} = 60 V dc; - P_T = 800 mW for 2N3019 and 2N3019S; P_T = 400 mW for 2N3057A, 2N3700UB; P_T = 500 mW for 2N3700 at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on device shall be permitted.

4.4.3.2 Group C inspection, JAN, JANTX, and JANTXV. Group C inspection (JAN, JANTX, and JANTXV, package related tests) shall be as follows and shall include those package or case-related tests which are performed periodically. Group C test on each package family shall be performed on devices from each six months production of devices (based on inspection lot identification codes) for each assembly location and die attach method. Electrical measurements (end points) and delta requirements shall be in accordance with the steps of table II herein as specified in the notes for table II. The sample size for each of these subgroups is: 22 devices, c = 0. Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

Step	Method	Condition
1	2066	
2	1051	Test condition C.
3	2036	Test condition E; not applicable for UB devices.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.3.4 Product acceptable for delivery. The product which may be accepted for delivery based a successful completion of group C testing shall be product of the particular package type with lot inspection identification codes of the 52 successive weeks beginning with the inspection lot identification code of the successful group C sample (i.e. the inspection lot identification codes accepted for delivery based on successful group C testing will include those sealed during the approximate 10 weeks during which the sample was being screened and tested plus those sealed in the 42 weeks following successful completion of group C testing making a total of 52 successive calendar weeks.) A manufacturer shall not accept product for delivery until after successful completion of the group C testing on product of the applicable package type with an inspection lot identification code less than 51 weeks prior to the inspection identification code of the product being considered for acceptance.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Collector-base time constant. This parameter may be determined by applying an rf signal voltage of 1.0 volt (rms) across the collector-base terminals, and measuring the ac voltage drop (V_{eb}) with a high- impedance rf voltmeter across the emitter-base terminals. With f = 79.8 MHz used for the 1.0 volt signal, the following computation applies:

$$r'_b \, , C_{c(ps)} = 2 \times V_{eb} \text{ (millivolts)}$$

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Solderability 1/ 2/	2026	15 leads, c = 0				
Resistance to solvents 2/ 3/ 4/	1022	15 devices, c = 0				
Subgroup 2						
Collector to base breakdown voltage	3001	Bias condition D; I_C = 100 µA dc	$V_{(BR)CBO}$	140		V dc
Emitter to base breakdown voltage	3026	Bias condition D; I_E = 100 µA dc	$V_{(BR)EBO}$	7		V dc
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 30 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES1}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		10	nA dc
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE1}	100	300	
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE2}	50	200	
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE3}	90		
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE4}	50	200	
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE5}	15		
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$ 1/		0.2	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B = 50 mA dc pulsed (see 4.5.1)	$V_{CE(sat)2}$ 1/		0.5	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$ 1/		1.1	V dc
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES2}		10	µA dc
Low-temperature operation		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE6}	40		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc; I_C = 1 mA dc f = 1 kHz	h_{fe}	80	400	
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 20 MHz	$\lvert h_{fe} \rvert$	5	20	
Input capacitance (output open circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		60	pF
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		12	pF
Noise figure	3246	V_{CE} = 10 V dc; I_C = 100 µA dc; R_g = 1 kΩ; Power bandwidth = 200 Hz	NF		4	dB
Collector to base time constant		V_{CB} = 10 V dc; I_C = 10 mA dc; f = 79.8 MHz (see 4.5.2)	$r'_b C_c$		400	ps
Pulse response		See figure 7	$t_{on} + t_{off}$		30	ns

See footnote at end of table.

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TABLE I. Group A inspection. - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = 25° C; t = 10 ms, 1 cycle (see figure 8)				
Test 1 2N3019, 2N3019S 2N3057A, 2N3700, 2N3700UB		V_{CE} = 10 V dc; I_C = 500 mA dc I_C = 180 mA dc				
Test 2 2N3019, 2N3019S 2N3057A, 2N3700, 2N3700UB		V_{CE} = 40 V dc; I_C = 125 mA dc I_C = 45 mA dc				
Test 3 2N3019, 2N3019S 2N3057A, 2N3700, 2N3700UB		V_{CE} = 80 V dc; I_C = 60 mA dc I_C = 22.5 mA dc				
Electrical measurements		See table II, steps 2, 4, and 7.				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ Separate samples may be used.
3/ Not required for laser marked devices.
4/ Not required for JANS devices.

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TABLE II. Groups B and C electrical measurements. 2/, 3/, 4/, 5/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES1}		10	nA dc
2.	Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES1}		20	nA dc
3.	Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		10	nA dc
4.	Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$ 1/		0.20	V dc
5.	Base emitter voltage (saturated)	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$ 1/		1.1	V dc
6.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE2}	50	200	
7.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc	h_{FE1}	100	300	
8.	Collector-emitter cutoff current	3041	Bias condition D; V_{CE} = 90 V dc; pulsed (see 4.5.1)	ΔI_{CES1}	±100% of initial value or 8 nA dc, whichever is greater.		
9.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc; pulsed (see 4.5.1)	Δh_{FE2}	±25% change from initial reading		
10.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	Δh_{FE1}	±25% change from initial reading		
11.	Collector-emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$\Delta V_{CE(sat)1}$ 1/	±50% mV dc change from previous measured value.		

See footnotes at end of table.

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TABLE II. Groups B and C electrical measurements - Continued. 2/, 3/, 4/, 5/

1/ Measured within 0.125 inch from the body of the device.

2/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 3, 4, 5, 6, and 7.

b. Subgroup 4, see table II herein, steps 2, 3, 4, 5, 6, 7, and 11.

c. Subgroup 5, see table II herein, steps 2, 3, 4, 5, 7, 8, and 10.

3/ The electrical measurements for group B, (JAN, JANTX, and JANTXV) 4.4.2.2 herein are as follows:

a. Steps 1, 2, and 4 of 4.4.2.2, see table II herein, steps 1, 8, 9, 10, and 11.

4/ The electrical measurements for table VII (for JANS only) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 3, 4, 5, 6, and 7 (JANS).

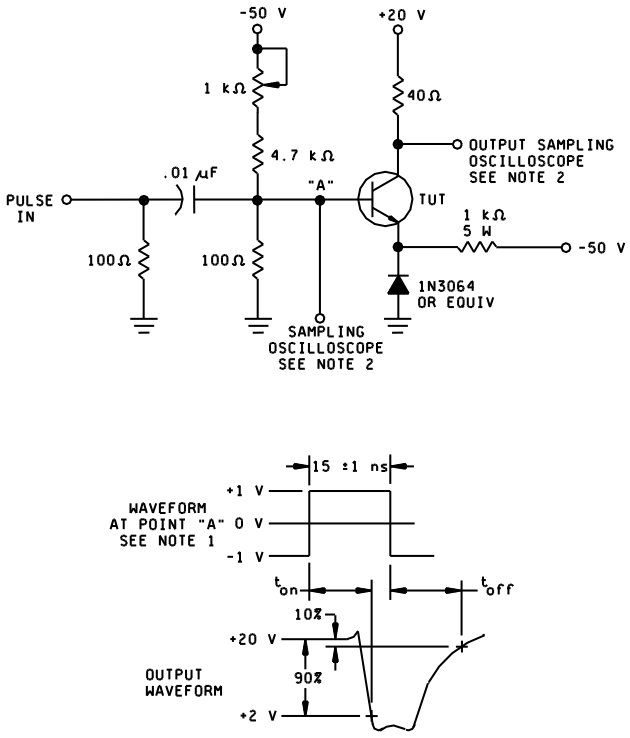
b. Subgroup 3, see table II herein, steps 1, 3, 4, 5, 6, and 7 (JANS).

c. Subgroup 6, see table II herein, steps 1, 3, 4, 5, 6, 7, 8, and 9 (JANS).

5/ The electrical measurements for group C, (JAN, JANTX, and JANTXV only) 4.4.3.2 herein are as follows:

a. Step 2 of 4.4.3.2, see table II herein, all steps.

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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2% and the generator source impedance shall be 50 ohms.

2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 12 pF, rise time ≤ 2.0 ns.

FIGURE 7. Nonsaturated switching-time test circuit.

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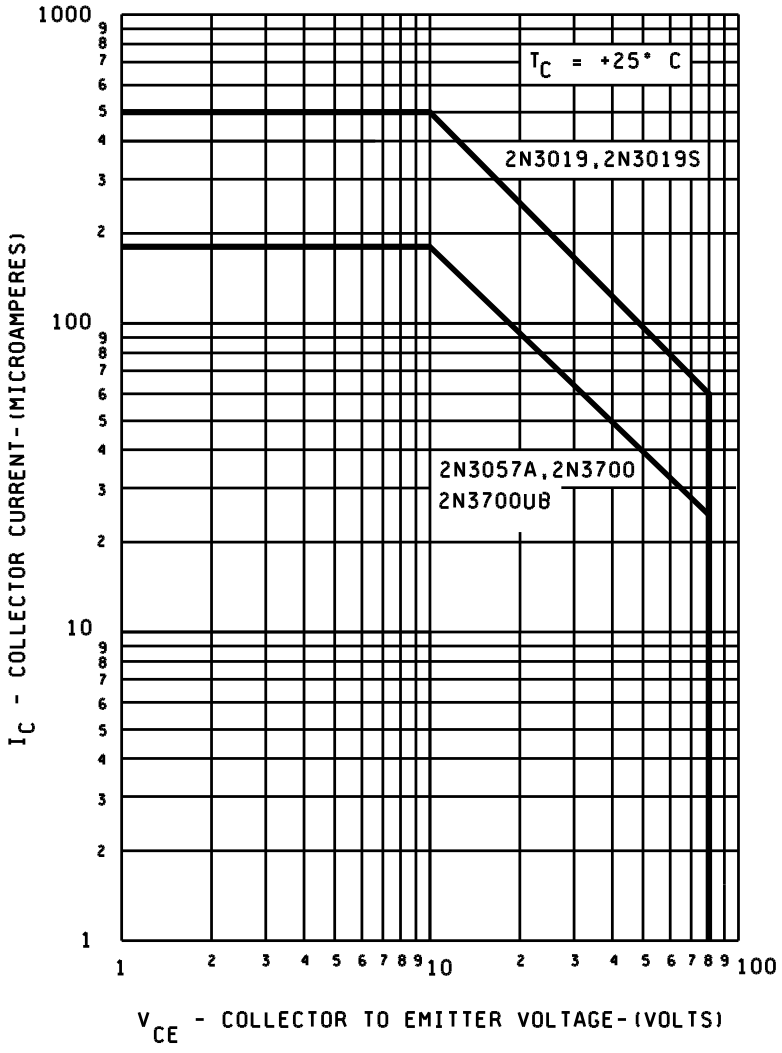


FIGURE 8. Maximum safe operating graph (DC).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17

Preparing activity:
DLA - CC

(Project 5961-1926)

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

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INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/391D	**2. DOCUMENT DATE (YYMMDD)** 97/12/09

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER TYPES 2N3019, 2N3019S, 2N3057A, 2N3700, AND 2N3700UB JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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